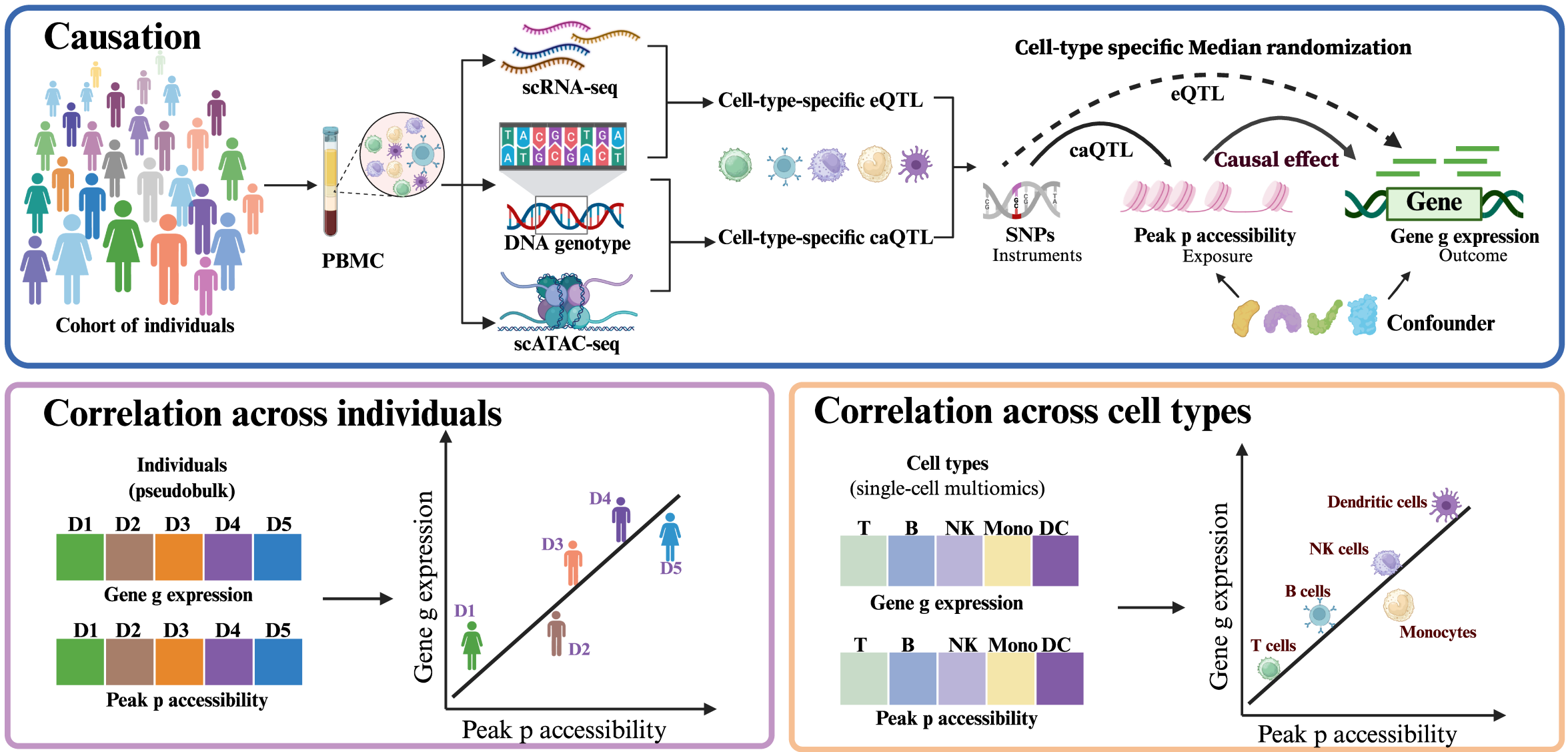
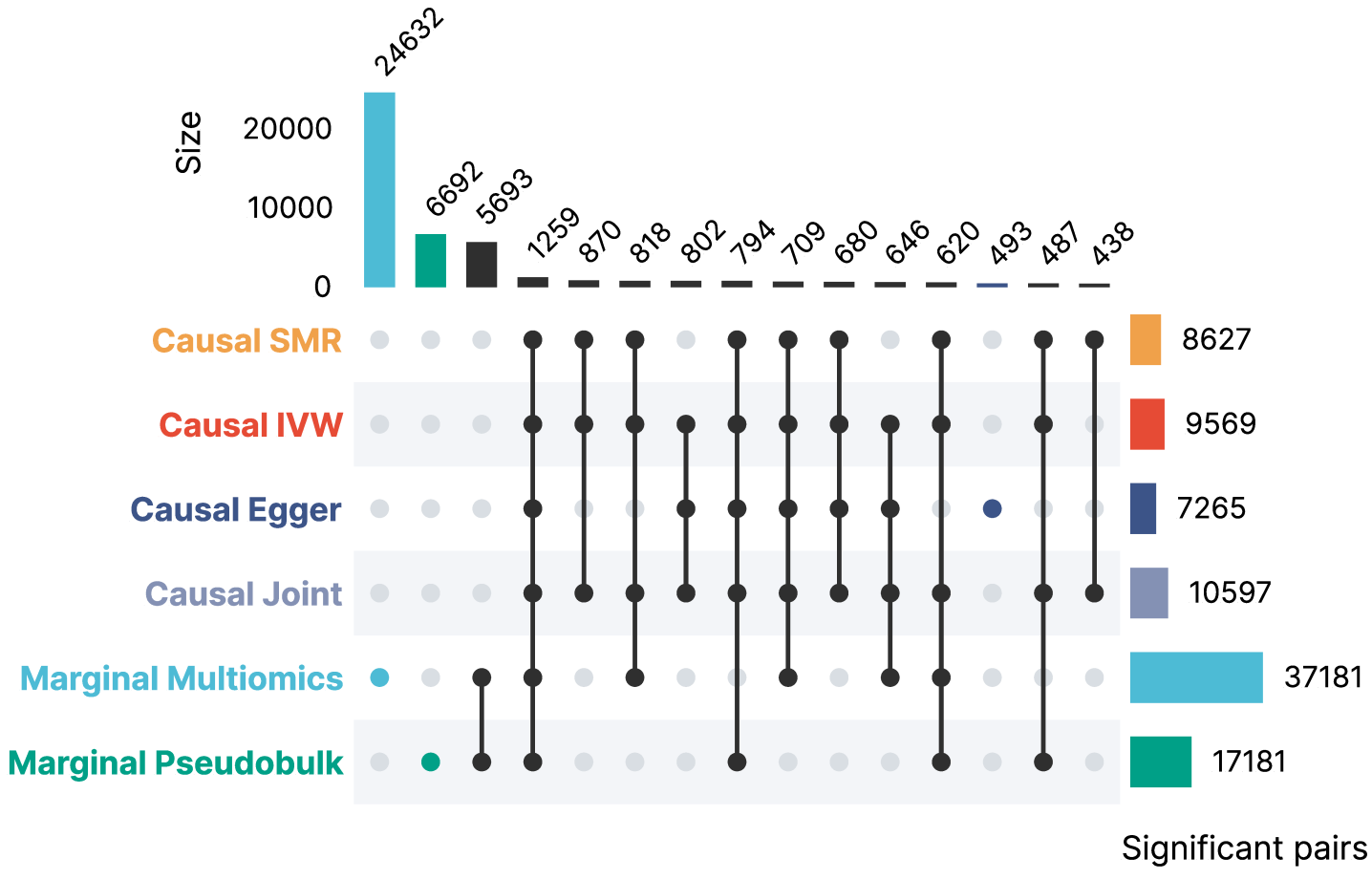
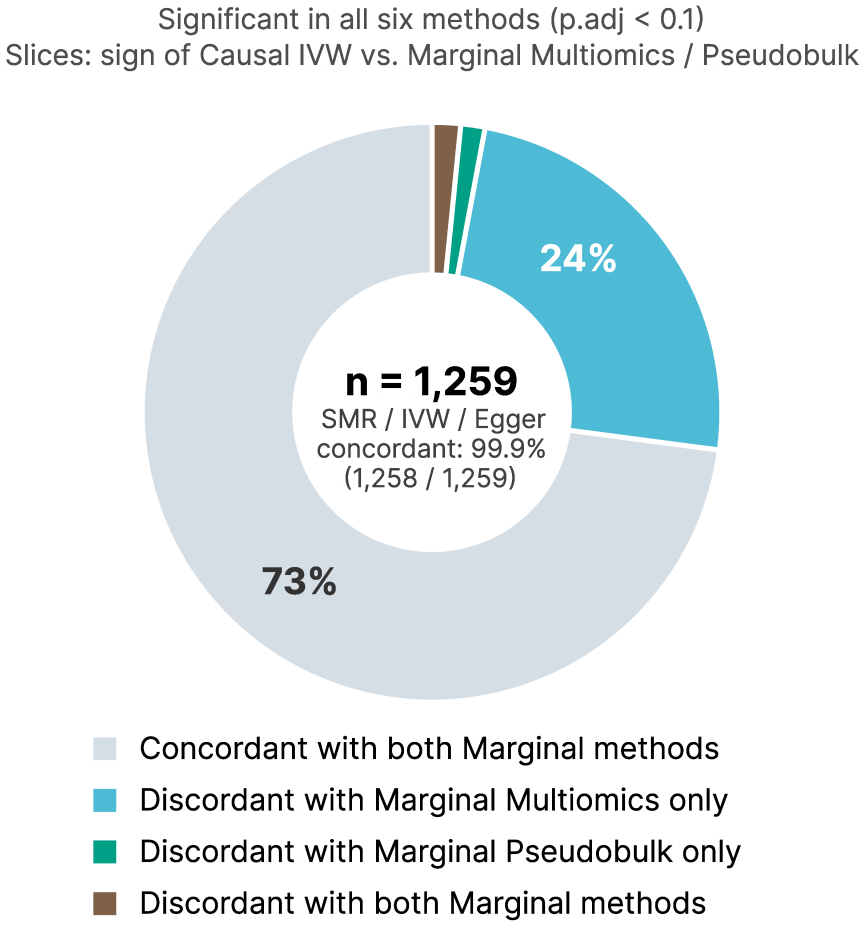
A


Causation
Cohort of individuals
PBMC
scRNA-seq
DNA genotype
scATAC-seq
Cell-type-specific eQTL
Cell-type-specific caQTL
Cell-type specific Median randomization
eQTL
caQTL
Causal effect
SNPs
Instruments
Peak p accessibility
Exposure
Gene
Gene g expression
Outcome
Confounder
Correlation across individuals
Individuals
(pseudobulk)
D1 D2 D3 D4 D5
Gene g expression
D1 D2 D3 D4 D5
Peak p accessibility
Gene g expression
Peak p accessibility
Correlation across cell types
Cell types
(single-cell multiomics)
T B NK Mono DC
Gene g expression
T B NK Mono DC
Peak p accessibility
Gene g expression
Peak p accessibility
Dendritic cells
NK cells
B cells
Monocytes
T cells

B


CD4 T cell
Size
20000
10000
0
24632
6692
5693
1259
870
818
802
794
709
680
646
620
493
487
438
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
8627
9569
7265
10597
37181
17181
Significant pairs

C


Effect direction
Significant in all six methods (p.adj < 0.1)
Slices: sign of Causal IVW vs. Marginal Multiomics / Pseudobulk
24%
73%
n = 1,259
SMR / IVW / Egger
concordant: 99.9%
(1,258 / 1,259)
Concordant with both Marginal methods
Discordant with Marginal Multiomics only
Discordant with Marginal Pseudobulk only
Discordant with both Marginal methods